

Meal Type clustering in Daily Food & Nutrition Dataset using K-Means Clustering

- *Modified the existing notebook [\(Link to Notebook\)](#) by changing the dataset and performing the following actions:*

Task	Actions Taken	Rationale	Difficulty Level (1-10)	Time Taken
Data Exploration & Preprocessing	Loaded dataset, removed unwanted columns, handled missing values, encoded categorical features, and removed outliers.	Ensuring data quality and preparing it for clustering.	5	45 min
Feature Engineering	Processed, scaled numerical attributes, and split data into train & test sets.	Standardization is necessary for distance-based clustering methods like K-Means.	4	30 min
Implemented K-Means Clustering	Tested different k values using the elbow method and Silhouette Score to find the optimal number of clusters.	Finding the best k value ensures meaningful cluster formation.	7	1 hour 30 min
Hyperparameter Tuning	Performed grid search over different k values, achieving the highest Silhouette Score (0.0839).	Fine-tuning k improves cluster separation.	6	50 min
Dimensionality Reduction (PCA)	Applied PCA and reduced dimensions to visualise clustering, observed drop in Silhouette Score (0.0407).	PCA helps in visualization but may impact cluster separation.	5	30 min

Model Evaluation	Compared results with different k values and PCA, analyzed cluster quality using Silhouette Score.	Evaluating model performance and cluster coherence.	4	30 min
Conclusion & Future Improvements	Identified suboptimal clustering, decided to explore DBSCAN for potential improvement.	Exploring more advanced clustering algorithms for better results.	3	15 min

1. Business Understanding

Objective:

The principal aim of the study is to use K-Means Clustering for meal types as classified by their nutritional values concerning the Daily Food & Nutrition Dataset. This classification should enable understanding dietary habits, optimize meal planning for individual users, and offer specific nutrition recommendations.

Key objectives include:

Meal classification: These should identify clusters of different types of meals such as breakfast, lunch, dinner, and snack by the nutritional profile.

2. Dataset Overview:

Dataset Used: *Daily Food & Nutrition Dataset*

<https://www.kaggle.com/datasets/adilshamim8/daily-food-and-nutrition-dataset>

Description: This dataset contains **10,000** records of **food intake data** from **1,000 unique users**, covering a wide variety of **food items** and their **nutritional values**.

3. Preparing Data

The following EDA steps are followed:

1. Load dataset.
2. Drop unwanted columns.
3. Handle missing values.

4. Encode categorical features.
5. Process datetime features (NA)
6. Remove outlier
7. Split into train & test sets.

4. Modelling:

Model Hyperparameters:

Number of clusters (k): has to explore for Maximum **Silhouette Score**.

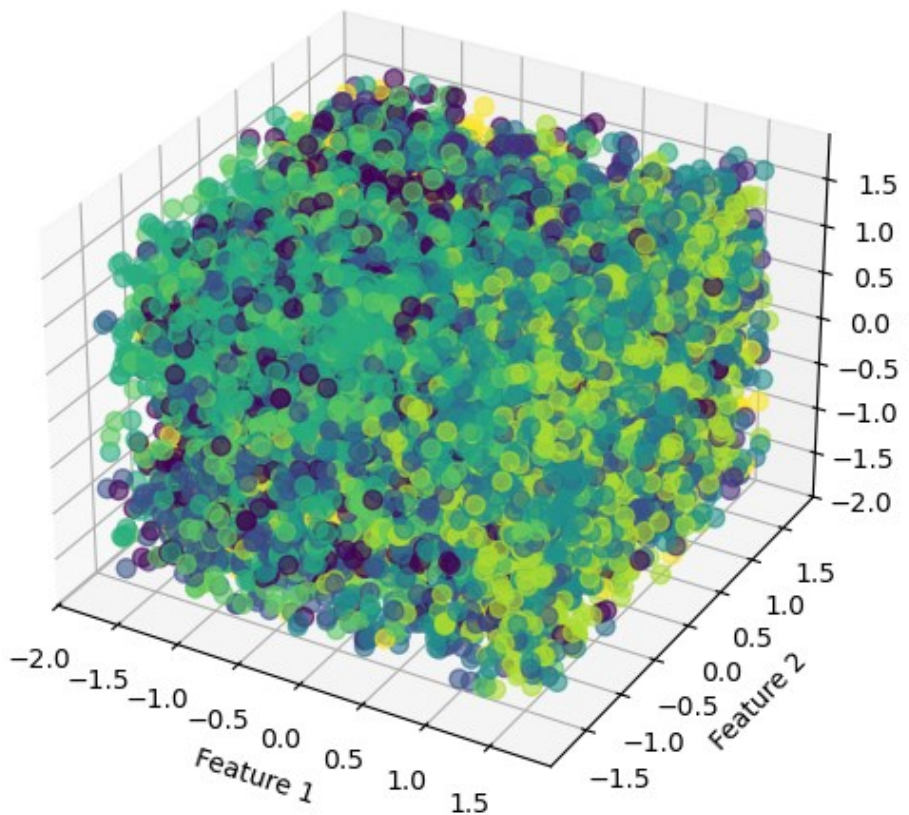
The K-Means assigns a cluster label (0-11) to each **data point** based on the **closest centroid**.

5. Evaluation:

Evaluation Metrics:

Best **K** (optimal clusters) according to **Silhouette Score (0.08272176061074789): 2**

K-Means Clusters in 3D



6. Observation:

- With hyperparameter tuning (**k value grid search**), we are able get **Silhouette Score of 0.0839295427364303**
- After **dimensionality reduction**, we are able get **Silhouette Score (after PCA): 0.04073845946537742**
- So, I am planning to explore more advanced algorithm like DBSCAN to get better result.